Question	Answer	Marks	Guidance
10(a)	energy required to separate (all) the nucleons (in the nucleus)	M1	**Allow** 'energy released when nucleons come together (to form nucleus)'. **Allow** 'work done' for 'energy required'. **Allow** 'protons and neutrons' for 'nucleons'. **Do not allow** any impression of a single nucleon.
	to infinity	A1	**Expect** 'from infinity' if the alternative M1 version of the M1 mark is given.
10(b)(i)	positron	B1	**Allow** 'beta +'. 'Beta' is incomplete. 'Beta–' is incorrect. **Ignore** symbols.
10(b)(ii)	$E = (\Delta)mc^2$	C1	
	$= 0.030377 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $= 4.54 \times 10^{-12}$ J	A1	Full substitution and answer needed. Unit not required but any unit given must be correct.
10(c)(i)	$F = L / 4\pi d^2$	C1	
	$= (7.98 \times 10^{27}) / (4\pi \times (8.15 \times 10^{16})^2)$ $= 9.56 \times 10^{-8}$ W m^{-2}	A1	Correct to at least 3 significant figures. AFC.

Question	Answer	Marks	Guidance
10(c)(ii)	number of He-4 nuclei formed per unit time $= (7.98 \times 10^{27}) / (4.54 \times 10^{-12})$	C1	The three C1 marks are all independent. No ECF from **10(b)(ii)**. Value $1.76 \times 10^{39}\ s^{-1}$ implies C1.
	number of H-1 nuclei fused = 4 × number of He-4 nuclei produced	C1	Value $7.03 \times 10^{39}\ s^{-1}$ implies first two C1 marks.
	mass of H-1 $= 1.67 \times 10^{-27} \times$ number of H-1 nuclei	C1	**Allow** use of 1.66×10^{-27}. Inclusion of 0.030377 in the mass calculation is XP.
	mass rate = $(4 \times 1.67 \times 10^{-27} \times 7.98 \times 10^{27}) / (4.54 \times 10^{-12})$ $= 1.17 \times 10^{13}\ kg\ s^{-1}$	A1	This line implies all three C1 marks. Correct to at least 3 significant figures. AFC.